Team Control Number

15041

Problem Chosen

A

2024

HiMCM/MidMCM

Summary Sheet

The International Olympic Committee (IOC) is organizing the 2032 Summer Olympics in Brisbane, Australia, with the aim of keeping the Games relevant and appealing to a global audience. To help determine which Sports, Disciplines, and Events (SDEs) should be added or removed, our team developed a comprehensive model to assess SDEs against six IOC criteria: **Popularity and Accessibility, Gender Equity, Sustainability, Inclusivity, Relevance and Innovation, and Safety and Fair Play**.

We identified **27 quantitative and qualitative sub-criteria/factors**, such as ticket sales, digital media engagement, leadership representation, environmental policies, global reach, youth appeal, and safety measures like injury rates and anti-doping compliance. Additionally, some of these factors are specific to the host country, such as cultural relevance. These factors provided a robust and comprehensive basis for evaluating each SDE's alignment with the IOC's vision.

With these 27 factors, we developed a **Hierarchical Multi-Criteria Decision Model (H-MCDM)**. Our H-MCDM provides a structured, transparent, and adaptable framework for quantitatively assessing each sport against the IOC's criteria and accounting for global trends and host-specific factors. Our approach integrates **Analytic Hierarchy Process (AHP)** and **Technique for Order Preference by Similarity to Ideal Solution (TOPSIS)** within a hierarchical structure. AHP is used at the top level to determine weights to the six main criteria based on their importance. At the sub-criteria level, predefined weights capture the specific relevance of each factor. TOPSIS is then applied sequentially, first to assess SDE performance at the sub-criteria level, then to aggregate scores at the main criteria and it then gives out an **overall score** to comprehensively measure an SDE's alignment with IOC criteria. The higher the score, the better the alignment with IOC criteria. A comprehensive ranking of SDEs can then be generated.

We tested our model on **eleven SDEs** with a known status for the three Olympics: Tokyo 2020, Paris 2024, and LA 2028. We chose **a diverse range of SDEs**, including eight sports that have been recently added or removed (Skateboarding, Sport Climbing, Breaking, Karate, Flag Football, Lacrosse Sixes, Coastal Rowing, and Baseball/Softball), as well as three that have been in the Olympic program since the 1988 Games or earlier (Athletics, Swimming, and Gymnastics). Athletics, Gymnastics, and Swimming consistently ranked highest, reaffirming their global appeal and historical significance. The model effectively adapts to varying cultural, regional, and strategic priorities of different host cities. For example, our model assigned karate a relatively high score of 0.5607 and a high rank of #4 for Tokyo 2020. However, for Paris 2024 and LA 2028, the model outputted scores of 0.3369 and 0.2621 with ranks of #10 and #11, respectively. The alignment of high-ranked sports with actual Olympic decisions validates our model's criteria and weighting approach.

We then applied our model to evaluate the suitability of the eight recently added or removed SDEs for the **2032 Brisbane Olympics**. Apart from the core sports, the **top three ranked SDEs** are **Skateboarding, Breaking and Sport Climbing**. Therefore, we recommend that these three SDEs be considered for inclusion in the Brisbane 2032 Olympic Summer Games. For future inclusion in the year **2036 and beyond**, we suggest **Padel, Pickleball** and **VR-archery** based on international growth and increasing popularity.

We conducted **sensitivity analyses** by adjusting the weights ofthe main criteria by ±10% and ±50%. The model remained stable, with core sports maintaining top rankings and recommended sports consistently scoring highly. It also shows that **Popularity and Accessibility** and **Relevance and Innovation** are the most influential criteria, ensuring the model reflects IOC strategic priorities. Lower-weighted criteria like Sustainability and Safety and Fair Play had a balanced impact, confirming the model's reliability across various scenarios. In conclusion, our H-MCDM offers a **reliable**, **data-driven** approach for evaluating SDEs and supports informed decision-making that aligns with the IOC's evolving vision

Dear Members of the International Olympic Committee,

We are pleased to present our evaluation of potential additions and modifications to the Olympic program for the 2032 Summer Games, focusing on how sports align with the six IOC Criteria for Sports Inclusion: *Popularity and Accessibility, Gender Equity, Sustainability, Inclusivity, Relevance and Innovation, and Safety and Fair Play*. Using a structured and data-driven approach, we assessed Sports-Driven Events (SDEs) that have been recently added, re-added, or removed based on historical data, current participation trends, audience engagement, and alignment with these criteria. Our model predicts that the first SDE to be considered for inclusion in Brisbane 2032 is Skateboarding, followed by Sport Climbing, and then Breaking.

To conduct our evaluation, we first defined each of the six criteria through 2-6 specific factors, totaling 27 factors across all criteria. Some of these factors are dependent on the host city, such as *cultural relevance*. Using these factors, we constructed a Hierarchical Multi-Criteria Decision Model, combining the Analytical Hierarchy Process (AHP) to weigh the importance of each criterion and the Technique for Order Preference by Similarity to Ideal Solution (TOPSIS) to evaluate each SDE's overall alignment with the IOC Criteria for Sports Inclusion. This approach ensures a comprehensive and context-sensitive evaluation. Our model gives each SDE a score on a scale of 0 to 1 for its alignment with each of the six criteria, and these scores are then aggregated into a final score, with a higher score indicating better alignment. The ranks of all considered SDEs are based on these final scores.

We then validated our model with 11 diverse SDEs, including eight sports that have been recently added or removed (Skateboarding, Sport Climbing, Breaking, Karate, Flag Football, Lacrosse Sixes, Coastal Rowing, and Baseball/Softball), as well as three core Olympic sports that have been included since the 1988 Games or earlier (Athletics, Swimming, and Gymnastics). We tested the model with data for different host cities, including Tokyo 2020, Paris 2024, and LA 2028. The three core SDEs consistently ranked highest for these Olympics, while the newer or less stable SDEs fluctuated in ranking depending on the host city. For example, Karate ranked high (4th) for Tokyo 2020 but very low for Paris 2024 (10th) and LA 2028 (11th), consistent with its removal from those Games. Ultimately, the recently added sports had higher rankings compared to those recently removed, suggesting the validity of our model. We also performed a sensitivity analysis, demonstrating that our model provides robust rankings.

We applied our validated model to the Brisbane 2032 data to evaluate whether the eight recently added or removed SDEs should remain or return. The top three ranked SDEs are Skateboarding, Sport Climbing, and Breaking. Therefore, we recommend these three SDEs be included in the Brisbane 2032 Olympic Summer Games.

Thank you for the opportunity to share our insights. We are confident that these recommendations will contribute to the continued success of the Olympic Games. If you have any questions or would like further details, please do not hesitate to contact us.

Sincerely,
HiMCM Olympic Consultants Team #15041

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1 Introduction

1.1 Problem Background

The Olympic Games is the largest multi-sport event in the world, and the International Olympic Committee (IOC) faces the continuous challenge of ensuring that they stay relevant, inclusive, and aligned with modern values. Over time, this has involved adding, removing, or reintroducing Sports, Disciplines, and Events (SDEs) to reflect changing cultural and global trends.

To guide these decisions, the IOC has established six criteria for Sports Inclusion: Popularity and Accessibility, Gender Equity, Sustainability, Inclusivity, Relevance and Innovation, and Safety and Fair Play. However, applying these criteria consistently and quantitatively across a wide range of sports is a complex task. Each SDE has unique cultural, logistical, and historical factors that need to be considered, making it difficult to evaluate them.

In this paper, we aim to address this challenge by developing a mathematical model capable of quantitatively assessing SDEs based on their alignment with the IOC's criteria for Sports Inclusion.

1.2 Restatement of the Problem

Specifically, we need to complete the following tasks:

- Identify and list factors that need to be considered for each of the six IOC's criteria for Sports Inclusion.
- Construct a mathematical model to evaluate how well a specific SDE aligns with IOC criteria for Sports Inclusion using identified factors.
- Validate the proposed mathematical model using at least three SDEs that have been added or removed from recent Olympic Games as well as at least three SDEs that have been in the Olympic program for an extended period of time.
- Predict the top three SDEs among recently added or removed ones to be considered for the 2032 Summer Olympics.
- Perform a sensitivity analysis to test the robustness of our proposed model and present its strengths and weaknesses.
- Draft a one- to two-page non-technical letter to the IOC that describes our findings and results

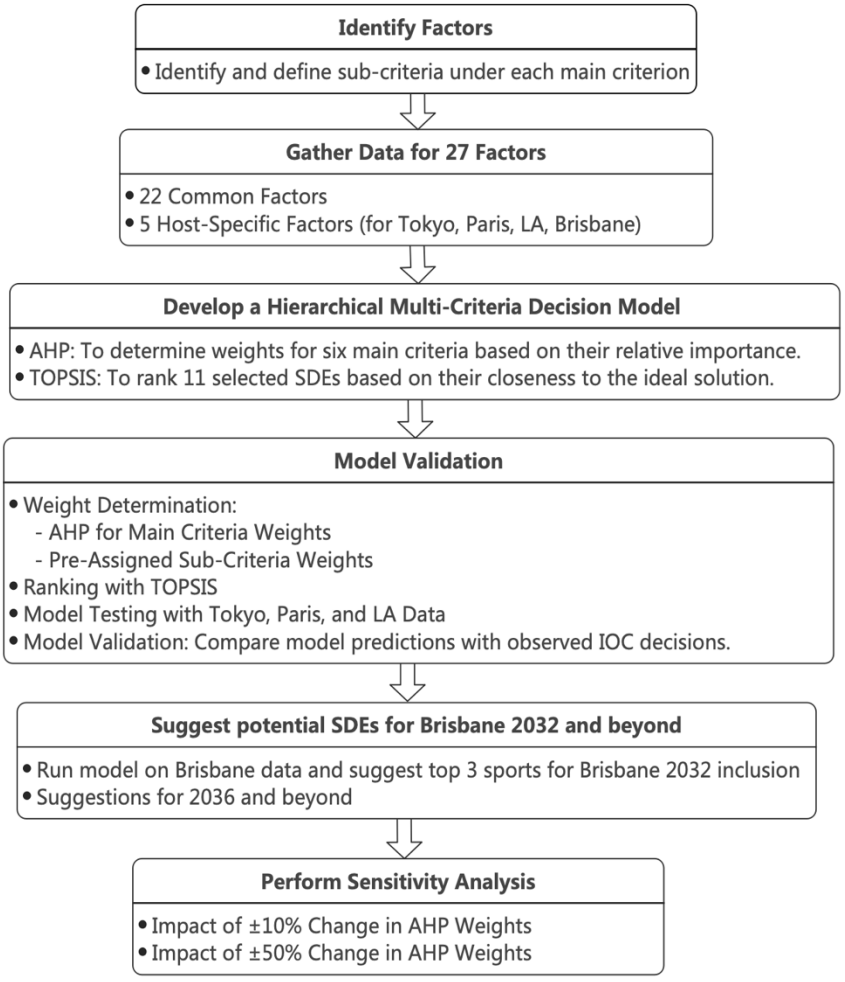


Figure 1: Our Work

1.3 Our work

Figure 1 illustrates the complete process of our work for all the tasks.

2 Assumptions and Justifications

To simplify the problem of quantitatively evaluating SDEs in Olympic games, we make the following assumptions.

Assumption 1: Venue Capacity as a Proxy for Ticket Sales and Attendance
In the absence of specific attendance figures, we assume that ticket sales are approximately equal to the venue's capacity.
Justification: Major sporting events, particularly high-profile games or finals, tend to attract audiences that fill the venue to its maximum capacity.
Assumption 2: Using 2023 World Championship Venue Capacity for Ticket Sales and Attendance Figures
We assume that the 2023 World Championship attendance data is the most recent and relevant.
Justification: Many sports hold their World Championships every two years, meaning 2023 is the latest available data until 2025. The 2023 championships provide the most current attendance information, reflecting recent audience interest and ticket sales.
Assumption 3: Estimating Attendance for last Coastal Rowing World Championship
For Coastal Rowing, where there is no ticketed venue, we assume an estimated attendance of 5,000 spectators.
Justification: Coastal rowing events are open to public viewing along shorelines, often attracting thousands of informal spectators. Based on similar outdoor events, an estimated value of 5,000 is a reasonable approximation for attendance.
Assumption 4: Using YouTube View Data as a Proxy for Digital Media Engagement
We assume that YouTube metrics (e.g., views) for each sport's 2023 World Championship can be used as a reliable indicator of overall digital media engagement.
Justification: YouTube is a major platform for sports content consumption, especially for large-scale international events. Metrics like views provide valuable insights into audience interest and popularity.
Assumption 5: Equal Events and Participation
We assume that all sports considered in this analysis will either maintain or transition toward gender equality in events and participation, in line with the IOC's commitment to gender equality.
Justification: Current Olympic sports have increasingly demonstrated gender balance in recent Games. For sports not yet included in the Olympics, we assume that compliance with IOC policies will ensure gender parity in participation and events.

3 Defining Criteria, Sub-Criteria, and Factors for Olympic SDE Selection

3.1 Trends and Insights from Recent Olympics

Our team conducted extensive research on the Olympic Games to identify trends and priorities in recent decisions on Sports, Disciplines, and Events (SDEs). We identified some major themes in recent Olympics, including:

- **Youth Relevance:** A growing emphasis on youth relevance to engage younger generations.
- **Urban and Modern Appeal:** A focus on urban and modern sports that appeal to audiences.

- **Host Country Interests:** Consideration of host country interests to ensure alignment with local culture and resources.

We also discovered two official IOC documents that were fundamental in guiding our selection of evaluation factors: the *Evaluation Criteria for Sports and Disciplines (2012)* [1] and the *Olympic Programme - Host City Proposal - Evaluation Criteria for Events* [2]. These trends and documents helped us select the sub-criteria and factors under each of six IOC criteria for Sports Inclusion as described in the next section, ensuring our evaluation aligns with both current Olympic trends and official IOC priorities.

3.2 Factors Considered in Evaluating IOC Criteria

This section describes the factors associated with the six IOC criteria used to evaluate SDEs in our model. Each factor is presented by its type, units/scale, and justification, as outlined in Table 1. A graphical display of the hierarchical relationship among the criteria, sub-criteria, and factors can also be found in Figure 3.

Table 1: Criteria, Sub-Criteria, and Factors for SDE Evaluation

Criteria & Sub-Criteria & Factors	Type	Units/Scale	Justification
Criteria #1: Popularity and Accessibility			
Global Popularity			Measures sport's appeal and interest.
- Ticket Sales and Attendance (last World Championship)	Quantitative	Count (tickets sold, attendees)	Reflects public interest in the sport through attendance at major events.
- Digital Media Engagement (last World Championship)	Quantitative	Social media metrics (views)	Measures social media and online platform popularity (e.g., Facebook, YouTube, Twitter)
Participation of Top Athletes	Qualitative	Presence of elite athletes (Yes/No)	High-profile athletes attract viewers, adding competitive value and increasing audience engagement.
Host Country Interest **	Qualitative	1–5 (1: Very low, 5: Very high)	High local interest enhances support and engagement within the host country.
Cost and Complexity			Ensures manageable costs and athlete capacity.
- Infrastructure Cost (Availability of Venues) **	Qualitative	Cost Level (Low, Moderate, High)	Infrastructure requirements should be minimal and cost-effective. Lower infrastructure costs make the sport more feasible for inclusion, as it can utilize existing facilities without requiring expensive upgrades.
- Athlete Quota	Quantitative	Athlete count	Assesses the number of athletes required for the sport, ensuring it fits within the IOC's total athlete quota [3]. A lower athlete count makes the sport easier to accommodate, increasing its feasibility for inclusion in the Olympic Games.
Governance	Qualitative	Presence of IOC-recognized IF (Yes/No)	The sport must be governed by an IF that is recognized by the IOC.

Historical Significance	Quantitative	Count (number of past Olympic Games)	Reflects the legacy and tradition of the sport, contributing to credibility, established audience interest, and enhancing the Olympic narrative.
Criteria #2: Gender Equity			
IF Leadership Gender Balance	Qualitative	Female Representation Ratio (Percentage)	Gender equality in the Executive Board (or highest decision-making body) of the IF. Promotes equal decision-making in sport governance.
Women and Sport Commission	Qualitative	Existence of commission (Yes/No)	Existence of a Women and Sport Commission and representation in decision-making body. Supports gender equity in governance.
Equal Events and Participation	Quantitative	Ratio (men to women)	The sport should aim for 50% female participation across Games, discipline, and event levels, where possible.
Criteria #3: Sustainability			
Environmental Policies	Qualitative	Compliance Level (None, Partial, Full)	Measures the existence and comprehensiveness of environmental policies adopted by the sport
Sustainable Sport & Event Toolkit (SSET)	Qualitative	Compliance Level (None, Partial, Full)	Use of any evaluation criteria such as the Sustainable Sport & Event Toolkit (SSET).
Social Responsibility **	Qualitative	Long-term impact Level (Low, Medium, High)	Evaluates how venues will be used after the Games and the long-term benefits for host cities. This includes community use, economic growth, and support for sustainability goals.
Infrastructure Cost (Availability of Venues) **	Qualitative	Cost Level (Low, Moderate, High)	Using existing venues reduces the need for new construction, which helps lower the carbon footprint and environmental impact. It also aligns with the IOC's sustainability goals by cutting resource use and emissions, making the event more eco-friendly.
Criteria #4: Inclusivity			
Global Reach (countries)	Quantitative	Count of countries that practice the sport.	The sport should be widely practiced by men in at least 75 countries and on four continents, and by women in at least 40 countries and on three continents.
Global Reach (continents)	Quantitative	Count of continents that practice the sport.	
IF Establishment	Quantitative	Years since the IF was founded	Supports global engagement with a stable IF.
IF Reach	Quantitative	IF member count	Represents the number of national federations affiliated with the IF, indicating the sport's international governance and organizational reach.
Continental Diversity	Quantitative	Count of continents with medaled nations	Measures how many continents have medal-winning nations, showing the sport's global reach and diversity.
National Championships	Quantitative	Percentage of National Federations organizing championships	Measures the formal structure and institutional support of the sport globally

Criteria #5: Relevance and Innovation			
Youth Relevance	Qualitative	Rating (1-5: 1 = No Relevance, 5 = Very High Relevance)	Assesses how well the sport appeals to younger audiences to keep the Olympics attractive to future generations.
Urban Relevance	Qualitative	Rating (1-5: 1 = No Relevance, 5 = Very High Relevance)	Measures how well the sport fits into urban settings.
Cultural Relevance **	Qualitative	Rating (1-5: 1 = No Relevance, 5 = Very High Relevance)	Evaluates the sport's importance in the cultural context of the host nation.
Innovation and Technology	Qualitative	Presence of Virtual Components (1: Low to 3: High)	Assesses the integration of modern elements like e-sports or augmented reality, reflecting current trends and technological advancements.
Criteria #6: Safety and Fair Play			
Athlete Health and Safety	Quantitative	Injury rate (number of injuries per 1,000 participants)	Evaluates the risk of injuries involved in the sport, ensuring that athlete health and safety are prioritized.
Anti-Doping and Fair Play Compliance	Qualitative	Compliance Level (Low, Moderate, High)	Comply with the Olympic Charter, the World Anti-Doping Code and the Olympic Movement Code on the Prevention of the Manipulation of Competition

Note that some factors are evaluated using a 'Smaller-is-Better' approach, meaning lower values are preferred. Specifically, **Infrastructure Cost**, **Athlete Quota**, and **Athlete Health and Safety - Injury Rate** are considered 'Smaller-is-Better' because lower values indicate greater feasibility or safety. For **Equal Events and Participation**, a ratio of 1:1 is ideal, because it is a balanced midpoint. All other factors follow a 'max-is-better' approach, where higher values are preferred. Additionally, several factors marked with red "**" in Table 1 are **host-specific**: **Host Country Interest, Infrastructure Cost (Availability of Venues), Social Responsibility, and Cultural Relevance**. Their values vary depending on the host, while the values for all other factors remain the same.

4 Hierarchical Multi-Criteria Decision Model for Olympic SDE Selection

To evaluate which SDEs align best with the IOC's criteria for inclusion in the Olympic Games, we developed a Hierarchical Multi-Criteria Decision (H-MCDM) Model. This model integrates the Analytic Hierarchy Process (AHP) for determining criteria weights and the Technique for Order Preference by Similarity to Ideal Solution (TOPSIS) for scoring and ranking the SDEs. The hierarchical structure allows for a systematic evaluation of multiple criteria and sub-criteria, ensuring a comprehensive analysis aligned with the IOC's vision (Figure 3). We will first describe how AHP and TOPSIS work and then present our proposed H-MCDM model.

4.1 Analytical Hierarchy Process (AHP) Model

The Analytic Hierarchy Process (AHP) was developed in the 1970s by Dr. Thomas Saaty to organize and analyze complex decisions [4]. Since there are multiple criteria to be considered, prioritizing them can be difficult. Instead, individuals use pairwise comparisons to quantify the relative importance of their criteria by completing a Pairwise

Comparison Matrix. This allows them to create weights for each criterion, thereby ranking them. The relative importance is defined on a scale of 1-9.

In the matrix, the intersection between row k and column j, $a_{k,j} = 1/a_{j,k}$, where $a_{k,j}$ shows how important criterion j is over criterion k. The following is an example of a three-factor Pairwise Comparison Matrix, X.

$$X = \begin{bmatrix} 1 & a & b \\ 1/a & 1 & c \\ 1/b & 1/c & 1 \end{bmatrix}$$

		Importance
Scale	1	Equal Importance
	3	Moderate importance
	5	Strong importance
	7	Very strong importance
	9	Extreme importance

Table 2: AHP Importance Scale

After this pairwise comparison matrix is obtained, the matrix needs to be normalized in order to interpret and give relative weights to each criterion. This normalization is made by dividing each value by the total column value using the following equation: $\bar{a}_{j,k} = a_{j,k} / \sum_{m=1}^{n} a_{m,k}$, where $\bar{a}_{j,k}$ is the normalized value of row j and column k in the normalized matrix and n is the number of criteria.

Using this normalized matrix, the relative importance of each criterion is then calculated by the Eigenvector method. At this point, when the criteria weights are produced, the produced weights should be checked for accuracy and have any data inconsistencies fixed. The consistency of the user's preferences needs to be checked. For example, if the users input that criterion A is more important than criterion B and criterion B is more important than criterion C, then it would be inconsistent for them to say criterion A is less important than criterion C. A perfectly consistent matrix would have inputs such that $a_{i,k} \times a_{k,j} = a_{i,j}$ for any criteria i, j, k.

The inconsistency of the pairwise comparison matrix can be measured using the consistency index (CI), which is based on the maximum eigenvalue λ_{max}. For a $n \times n$ comparison matrix, CI = (λ_{max} -n) / (n-1). In order to verify whether this consistency index is adequate, the consistency rate (CR), the ratio between the consistency index and the random consistency index (RI), is used. The matrix will be considered consistent if the resulting ratio is less than 10%, that is, $\mathrm{CR} = \mathrm{CI} / \mathrm{RI} < 10\%$.

If the matrix is inconsistent, we determine the most inconsistent comparison and reevaluate the decision. $a_{i,j} = p_i/p_j$ under consistent circumstances where p_i is the final weight for the ith criterion. Therefore, using the final weights p_i, what $a_{i,j}$ can be calculated and the inconsistency of an input $\varepsilon_{i,j}$ has the relationship $a_{i,j} = \varepsilon_{i,j}(p_i/p_j)$. The inconsistency for every input can then be determined, and we will reevaluate the input with the highest inconsistency.

4.2 Technique for Order Preference by Similarity to Ideal Solution (TOPSIS) Model

Technique for Order Preference by Similarity to Ideal Solution (TOPSIS) model was originally developed by Ching-Lai Hwang and Yoon in 1981 [5], and is a multi-criteria decision-making method. TOPSIS is based on the concept that the best solution has the shortest distance from the positive-ideal and the longest distance from the least ideal solution. Through calculating an overall index based on the distances form the ideal solutions, the alternatives can be ranked to help make final decision. We will describe this model step by step below.

Step 1: Establish a Performance Matrix

Assuming there are J criteria to evaluate I alternatives/options, then we can have a performance matrix,

$$X = \begin{bmatrix} x_{11} & \cdots & x_{1J} \\ \cdots & \cdots & \cdots \\ x_{I1} & \cdots & x_{IJ} \end{bmatrix}$$

Step 2: Calculate Normalized Matrix

In this procedure each performance rating x_{ij} in X is divided by its norm. The normalized ratings $y_{ij}(i = 1, 2, \dots, I; j = 1, 2, \dots, J)$ can be calculated with the following equation:

$$y_{ij} = \frac{x_{ij}}{\sqrt{\sum_{j=1}^{n} {x_{ij}}^2}}$$

Through this process, it makes comparison across attributes easier as there are dimensionless units. The normalized performance ratings y_{ij} can be given as a matrix Y as shown below:

$$Y = \begin{bmatrix} y_{11} & \cdots & y_{1J} \\ \cdots & \cdots & \cdots \\ y_{I1} & \cdots & y_{IJ} \end{bmatrix}$$

Stage 3: Calculate Weighted Normalized Matrix

The weighted and normalized performance rating $v_{ij}(i = 1, 2, \dots, I; j = 1, 2, \dots, J)$ is calculated by the following equation:$v_{ij} = W_j * y_{ij}; (i = 1, 2, \dots, I; j = 1, 2, \dots, J)$. Weights W_j can be manually assigned or based on AHP models. These weighted ratings are combined to form the weighted-normalized decision matrix V:

$$V = \begin{bmatrix} v_{11} & \cdots & v_{1J} \\ \cdots & \cdots & \cdots \\ v_{I1} & \cdots & v_{IJ} \end{bmatrix}$$

Step 4: Calculate the ideal best (positive ideal) and ideal worst (negative ideal)

A^* and A^- are denoted as the positive and negative ideal solution sets respectively, which can be determined from matrix V is:

$$A^* = \left[v_1^*, v_2^*, \dots, v_J^*\right], A^- = [v_1^-, v_2^-, \dots, v_J^-]$$

Where,

$$v_J^* = \begin{Bmatrix} max\ v_{ij}, if\ j\ is\ a\ benefit\ attribute \\ \min v_{ij}, if j\ is\ a\ cost\ attribute \end{Bmatrix}, v_J^- = \begin{Bmatrix} min\ v_{ij}, if\ j\ is\ a\ benefit\ attribute \\ \max v_{ij}, if j\ is\ a\ cost\ attribute \end{Bmatrix}.$$

Step 5: Calculate the separation measure

The separation measure is the distance of each alternative rating from both the positive and negative ideal solutions which is obtained by applying the Euclidean distance theory, shown by the following equations:

$$s_i^* = \sqrt{\sum_{j=1}^{J} (v_{ij} - v_j^*)^2}, \qquad s_i^- = \sqrt{\sum_{j=1}^{J} (v_{ij} - v_j^-)^2}$$

Step 6: Calculate the Performance Score, which measures the closeness to the ideal best

The overall performance score V_i for each alternative A_i is obtained from the following equation:

$$V_i = \frac{s_i^-}{s_i^- + s_i^*}$$

Step 7: Rank the preference order

The alternatives are ranked based on V_i values. A higher value suggests a better choice.

4.3 Our Hierarchical Multi-Criteria Decision Model

Our modeling process is described in Figure 2 while Figure 3 shows our Hierarchical Multi-Criteria Decision Model along with our hierarchical data structure in more details. More specifically, after we collect all factor-level data for each SDE, we first transform all factors' values, x_{ij}, for SDE i and factor j, so that higher values represent better performance: for 'Smaller-is-Better' factors j for SDE i, we use transformation $x'_{ij} = x_{max,j} - x_{ij}$; for 'Intermediate-is-Best' Indicators: Apply a function that peaks at the ideal value.

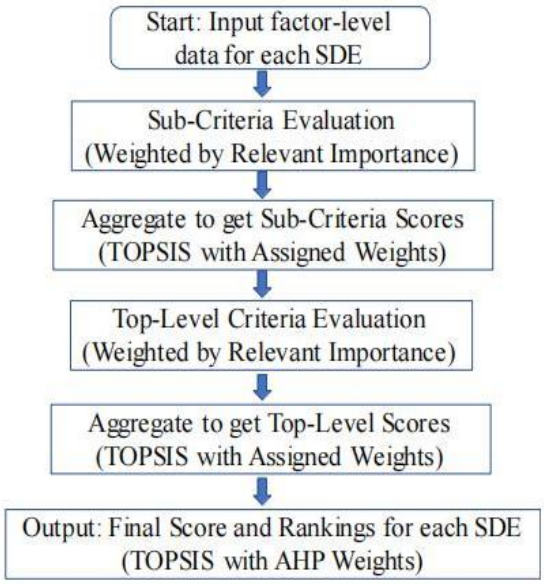


Figure 2: Workflow of our Hierarchical Multi-Criteria Decision Model

The final SDE score and ranking were obtained by applying TOPSIS sequentially from the bottom level (sub-criteria) to the top level (overall scoring). The sub-criteria-level score is calculated using TOPSIS with pre-assigned weights based on each factor's perceived importance. Then these sub-criteria-level score and other factors within a specific criterion was aggregated using another TOPSIS model with pre-assigned weights to get, $C_{i,k}$ represents the score of SDE i under main criterion k. The criteria-level scores, $C_{i,k}$, were aggregated using TOPSIS with weights from AHP to get final scores S_i for each SDE i, indicating their alignment with the IOC criteria. All S_i will be between 0 and 1 with higher scores indicating better alignment with the IOC criteria.

5 Model Validation

In this section we described how we validated our proposed H-MCDM model using selected SDEs and three Olympics (Tokyo 2020, Paris 2024 and LA 2028) where the status of each selected SDEs is known.

5.1 SDEs Selected for Testing

We select eleven SDEs for model testing, which covers a diverse range of sports, including three historical core sports: Athletics (Track and Field), Swimming, and Gymnastics (Artistic) and 8 recently Added, Removed, or Announced for Future Inclusion:

1. Skateboarding (Added in 2020, continued in 2024 and 2028)
2. Sport Climbing (Added in 2020, continued in 2024 and 2028)
3. Breaking (Added in 2024, will be removed in 2028)
4. Karate (Added in 2020, discontinued after that year)
5. Flag Football (To be added in 2028)
6. Lacrosse Sixes (To be added in 2028)
7. Coastal Rowing (To be added in 2028)
8. Baseball/Softball (Removed after 2008, reintroduced in 2020, removed again for 2024, and will return in 2028)

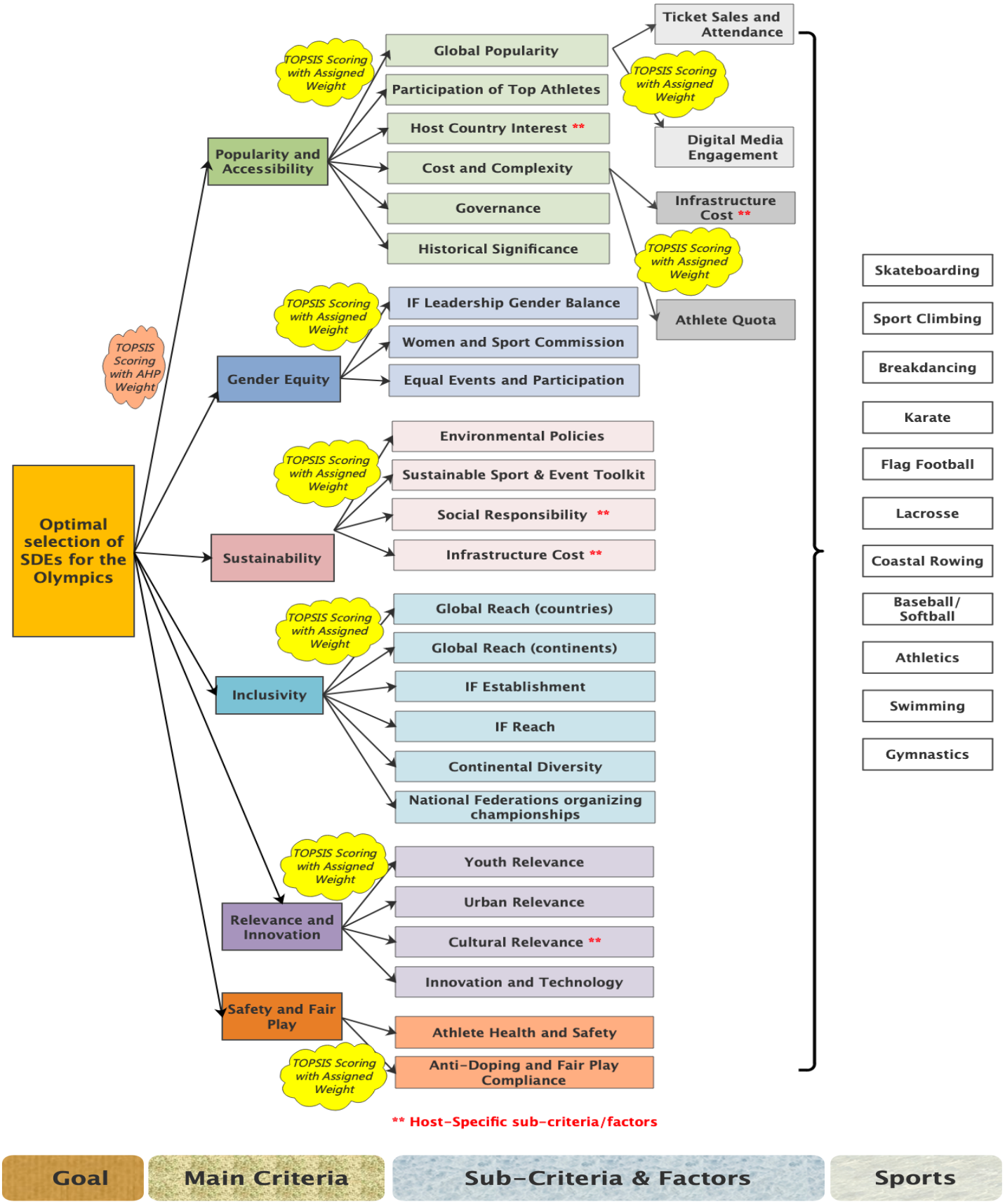


Figure 3: Decision-Making Hierarchy for Evaluating Olympic SDEs using AHP and TOPSIS

This selection includes a balanced variety of sports for assessing how well the model evaluates a range of SDEs across different criteria and validates the IOC's criteria for Olympic program inclusion.

- **New Sports with Youth Appeal**: Skateboarding, Sport Climbing, and Breaking are recently added sports with a strong youth appeal.
- **Culturally Significant Sports**: Karate and Lacrosse contribute cultural diversity, while Flag Football and Baseball/Softball add strong regional appeal.
- **Environmental Considerations**: Coastal Rowing uses sustainable practices and requires minimal infrastructure.
- **Core Olympic Values**: Athletics, Swimming, and Gymnastics are long-standing Olympic sports that meet high standards for inclusivity, popularity, and gender equity.

5.2 Data collection

This section explains our data collection methods for the 27 factors used in our model.

(1) Ticket Sales and Attendance (Last World Championship): Based on Assumptions 1 and 2, we used the 2023 World Championships' venue capacity as a proxy for ticket sales and attendance.

Table 3: Venue Capacity at 2023 World Championships for Eleven Sports

Sport	Event	Location	Venue Capacity
Skateboarding	World Skateboarding Tour Street World Championships 2023	Ariake Colosseum, Tokyo, Japan	10,000
Sport Climbing	IFSC Climbing World Championships 2023	PostFinance Arena, Bern, Switzerland	17,000
Breaking	World Breaking Championship 2023	Ladeuzeplein, Leuven, Belgium	2,500
Karate	World Karate Championships 2023	Papp László Budapest Sportaréna, Budapest, Hungary	12,500
Flag Football	IFAF Flag Football World Championship 2023	Charlotte, USA	14,000*
Lacrosse	World Lacrosse Championship 2023	Snapdragon Stadium, San Diego, USA	35,000
Coastal Rowing	World Rowing Coastal Championships 2023	Barletta, Italy	5,000**
Baseball/Softball	WBSC Baseball World Cup 2023	Taipei City and Taichung, Taiwan	20,000
Athletics	World Athletics Championships 2023	National Athletics Centre, Budapest, Hungary	36,000
Swimming	FINA World Aquatics Championships 2023	Marine Messe Fukuoka, Fukuoka, Japan	15,000
Gymnastics (Artistic)	FIG Artistic Gymnastics World Championships 2023	Sportpaleis, Antwerp, Belgium	23,000

*For Flag Football, missing data was estimated at 14,000 based on similar venues, such as the Royal Farms Arena in Baltimore.

**For Coastal Rowing, 'Open shoreline' attendance was estimated at 5,000 based on Assumption 3.

(2) Digital Media Engagement (Last World Championship): We use YouTube views as a proxy for digital media engagement, based on Assumption 4. The view counts for the 2023 World Championships are taken from YouTube sources [6-16].

(3) Participation of Top Athletes: All eleven sports had participation from top athletes in either recent Olympics or respective Championships [17-27].

(4) Host Country Interest:** *This factor is specific to the host country*. A score is assigned based on research and estimation of local interest levels, ranging from 1 (very low) to 5 (very high).

(5) Infrastructure Cost (Availability of Venues) **: *This factor is specific to the host country*. A score is assigned based on our research into the availability of existing facilities and an estimation of the infrastructure cost level, ranging from 1 (low) to 3 (high). This is a 'Smaller-is-Better' factor, indicating that a lower score is preferred for better feasibility.

(6) Estimated Athlete Quota: Numbers are estimated based on recent Olympic quotas. This is a 'Smaller-is-Better' factor, meaning that a lower number of athletes is preferred for greater feasibility, particularly in terms of re-source allocation and infrastructure use.

(7) Governance: All 11 sports have an IOC-recognized International Federation (IF), as confirmed by the *HiMCM_Olympic_Data.xlsx* file from HiMCM 2024 Problem A.

(8) Historical Significance: The count of past Olympic Games each sport has participated in is sourced from the HiMCM_Olympic_Data.xlsx file provided in HiMCM 2024 Problem A.

(9) IF Leadership Gender Balance: The female percentage in IF leadership is calculated from various sources [28-38]. A value of 50% is considered ideal, indicating balanced gender representation.

(10) Women and Sport Commission: All International Federations (IFs) have a Women and Sport Commission or an equivalent body, as confirmed through our research.

(11) Equal Events and Participation: All sports maintain or are expected to maintain gender equality in events and participation based on Assumption 5. A value of 50% is considered ideal.

(12) Environmental Policies: Values are assigned from 0 (None) to 1 (Full) based on research and estimation.

(13) Sustainable Sport and Event Toolkit (SSET): Compliance with the SSET is scored as 0 (None), 0.5 (Partial), or 1 (Full), based on our research.

(14) Social Responsibility: *This factor is specific to the host country*. Values are assigned on a scale from 1 (Low) to 3 (High), based on the estimated long-term environmental impact according to research.

(15) Infrastructure Cost (Availability of Venues): This factor is identical to factor (5), however it is used in evaluating the Sustainability criterion.

(16-21) Six Inclusivity Factors: Data collected for six Inclusivity factors is from sources [28-38]: Global Reach (countries), Global Reach (continents), IF Establishment, IF Reach (member count), Continental Diversity, and Percentage of National Federations organizing championships (%).

(22-24) Youth Relevance, Urban Relevance, Cultural Relevance: Scored from 1 (No Relevance) to 5 (Very High Relevance) based on our research and estimation of the sport's appeal to younger audiences, suitability for urban settings and assessment of the sport's cultural importance in the host nation. *Cultural Relevance is specific to the host country.*

(25) Innovation and Technology: Scored from 1 (Low) to 3 (High) based on research and estimation of the presence of virtual components.

(26) Injury Rate (per 1,000 participants): Injury rate data for each sport was collected from sources [39-49]. This is a 'Smaller-is-Better' factor, indicating that a lower rate is preferred.

(27) Anti-Doping & Fair Play Compliance: This factor is scored from 1 (low) to 3 (high), based on our research into the sport's level of compliance with anti-doping and fair play standards.

Table 4: Data collection for factors (Host city: Tokyo)

	Skate-boarding	Sport Climb-ing	Break-ing	Karate	Flag Football	Lacrosse	Coastal Rowing	Base-ball/Soft-ball	Athletics (Track and Field)	Swim-ming	Gym-nastics

Ticket Sales	10,000	17,000	2,500	12,500	14,000	35,000	5,000	20,000	36,000	15,000	23,000
Digital Media Engagement	612K	322K	2.3M	252K	271K	52K	1.2K	159K	353K	217K	360K
Top Athletes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Host Country Interest**	High	Moderate to High	Low	Very High	Low	Low	Low	Very High	High	High	Very High
Infrastructure Cost (Availability of Venues) **	Moderate	Moderate	Low	Low	Moderate	Moderate	Moderate	Moderate	High	High	Low
Estimated Athlete Quota	80	40	32	80	192	240	64	234	1900	1000	192
Governance	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Historical Significance	2	2	1	1	0	2	0	6	30	30	30
IF Leadership Balance (Female Percentage)	23%	27%	21%	13%	20%	53%	38.00%	29%	50%	38%	29%
Women and Sport Commission	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Equal Events and Participation	1:1	1:1	1:1	1:1	1:1	1:1	1:1	1:1	1:1	1:1	1:1
Environmental Policies	Partial	Full	None	None	Partial	Partial	Full	Partial	Full	Partial	Partial
Sustainable Sport and Event Toolkit	Partial	High	Low	Low	Partial	Partial	High	Partial	High	Partial	Partial
Social Responsibility**	High	Medium	Medium	Low	Low	Low	Medium	High	High	High	Medium
Infrastructure Cost (Availability of Venues)**	Moderate	Moderate	Low	Low	Moderate	Moderate	Moderate	Moderate	High	High	Low
Global Reach (countries)	~80	~90	~60	~190	~50	~60	~30	~200	~210	~200	~150
Global Reach (continents)	6	6	6	6	6	4	4	6	6	6	6
IF Establishment and Reach (Years Established)	100	17	67	54	26	16	132	11	112	116	143
IF Establishment and Reach (member count)	130	100	100	200	74	92	156	141	214	209	160
Continental Diversity (Medaled Continents Count)	5	3	4	5	2	2	3	3	6	4	4
National Championships (%)	60%	75%	55%	85%	40%	50%	65%	80%	95%	90%	85%
Youth Relevance	5	4	5	4	3	3	2	4	5	4	4
Urban Relevance	5	4	5	3	4	2	1	3	4	3	3

Cultural Relevance**	3	3	2	5	1	1	1	3	3	3	3
Innovation and Technology	Medium	Medium	Low	Low	Low	Low	Low	Medium	Medium	Low	Low
Injury Rate (per 1,000 participants)	8	3	4	88	5	5	1	3	4	1	8
Anti-Doping & Fair play compliance	Medium	High	Medium	High	Low	Medium	High	Medium	High	High	High

** This factor varies based on host country. Due to space limitation, we only report values for these factors specific to Tokyo for illustration here.

Data for all 27 factors were collected for **Tokyo** (see Table 4). Similar data was also gathered for **Paris, LA, and Brisbane**, with differences limited to **host-specific factors: Host Country Interest, Infrastructure Cost (Availability of Venues), Social Responsibility, and Cultural Relevance**. For these host-specific factors, adjustments were made based on research unique to each host city.

5.3 AHP calculated weight for final score

We used AHP to determine weights for our six major criteria: *Popularity and Accessibility, Gender Equity, Sustainability, Inclusivity, Relevance and Innovation, and Safety and Fair Play*. A pairwise comparison matrix was constructed accordingly, as shown below, based on our team's consensus to compare the importance of each criterion.

$$X = \begin{bmatrix} 1 & 4 & 4 & 3 & 2 & 5 \\ 1/4 & 1 & 1 & 1/2 & 1/3 & 2 \\ 1/4 & 1 & 1 & 1/2 & 1/3 & 2 \\ 1/3 & 2 & 2 & 1 & 1/2 & 3 \\ 1/2 & 3 & 3 & 2 & 1 & 4 \\ 1/5 & 1/2 & 1/2 & 1/3 & 1/4 & 1 \end{bmatrix}$$

We first performed a consistency check and calculated the Consistency Ratio (CR): which had a value of 0.0545. Since this $CR < 0.10$, the consistency of the comparison matrix is acceptable. The corresponding weights for each order are then obtained through AHP as shown in Table 5 and Figure 4. Popularity and Accessibility has highest weight because it reflects public interest and is a key factor in the financial success of the Olympics. Relevance and Innovation has the second highest weight because of the recent emphasis on youth appeal and urban relevance when selecting SDEs. During the construction of the AHP pairwise comparison matrix, we agreed to place more emphasis on these two criteria, and the resulting weights reflect our preferences.

Table 5: AHP calculated weights for 6 main criteria

Criterion	Weight
Popularity and Accessibility	0.3744
Gender Equity	0.0889
Sustainability	0.0889
Inclusivity	0.1509
Relevance and Innovation	0.2422
Safety and Fair Play	0.0544

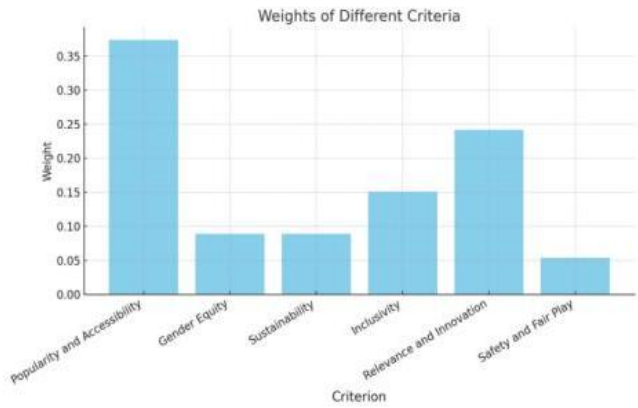


Figure 4: AHP calculated weights for 6 main criteria

5.4 Predefined weights for sub-criteria

Predefined weights for sub-criteria under each main criterion were determined by our team's consensus:

1. Global Popularity: Ticket Sales (0.9), Digital Engagement (0.1)
2. Cost & Complexity: Infrastructure Cost (0.9), Athlete Quota (0.1)
3. Popularity & Accessibility: Global Popularity (0.3), Top Athletes (0.1), Host Interest (0.4), Cost (0.1), Governance (0.05), Historical Significance (0.05)
4. Gender Equity: Leadership Balance (0.1), Women & Sport Commission (0.1), Equal Events & Participation (0.8)
5. Sustainability: Environmental Policies (0.1), SSET (0.1), Social Responsibility (0.4), Infrastructure Cost (0.4)
6. Inclusivity: Global Reach (Countries: 0.1, Continents: 0.1), IF Establishment (0.2), Member Count (0.2), Diversity (0.1), National Championships (0.3)
7. Relevance & Innovation: Youth Relevance (0.3), Urban Relevance(0.2), Cultural Relevance(0.4), Innovation(0.1)
8. Safety & Fair Play: Injury Rate (0.8), Anti-Doping (0.2)

5.5 Validation of our Hierarchical Multi-Criteria Decision Model

We applied our H-MCDM model to evaluate 11 SDEs across three Olympic hosts (Tokyo 2020, Paris 2024, LA 2028) as the status of 11 SDEs for these three Olympic is known. For each host city, a final score is generated for each of 11 SDEs. A higher score suggests better alignment with 6 IOC Criteria for Sports Inclusion.
Table 6 below shows the scores and ranks from our H-MCDM model. Here's what the results tell us:
Core Sports Consistency: Athletics, Gymnastics, and Swimming consistently rank in the top three, reflecting their global reach, accessibility, and historical importance, validating the model's alignment with IOC priorities.
Recently Added/Discontinued Sports:

- Karate: Our H-MCDM model ranked this sport high (rank #4) for Tokyo 2020 but very low for Paris 2024 (rank #10) and LA 2028 (rank #11), reflecting its popularity in Japan and cultural significance as a sport closely associated with the host country. This is consistent with the fact that Karate was added in Tokyo 2020 but removed from both Paris 2024 and LA 2028.
- Breaking: This sport was ranked highly for Paris 2024 (rank #4), but ranked much lower for Tokyo 2020 (rank #8) and LA 2028 (rank #10). This aligns with its urban appeal and cultural relevance for Paris.
- Flag Football, Lacrosse, Coastal Rowing and Baseball/Softball: These four sports are ranked higher for LA 2028 than Tokyo 2020 and Paris 2024, which is consistent with the fact that these four sports will be included for LA 2028.

Therefore, our model aligns closely with IOC decisions, ranking newly added sports like Karate (Tokyo), Breaking (Paris), and Baseball/Softball (Los Angeles) highly due to their host-specific cultural relevance. Consistent strong scores for Skateboarding and Sport Climbing highlight their global popularity and alignment with IOC goals for youth and innovation. Lower scores for newly included sports like Flag Football and Lacrosse reflect their need for greater global appeal. This demonstrates the model's robustness in capturing both global and host-specific priorities. The alignment of high-ranked sports with actual Olympic decisions (Tokyo 2020, Paris 2024, LA 2028) validates the model's **criteria and weighting** approach.

Table 6: Scores and ranks of 11 selected SDEs evaluated for three different host nations

SDE	Tokyo 2020		Paris 2024		Los Angeles 2028	
	Score	Rank	Score	Rank	Score	Rank
Athletics	0.8095	1	0.9685	1	0.9521	1
Gymnastics	0.7197	2	0.693	2	0.6438	2
Swimming	0.6269	3	0.5587	3	0.5701	3
Karate	0.5607	4	0.3369	10	0.2621	11
Baseball/Softball	0.5276	5	0.3527	9	0.4845	4
Skateboarding	0.5265	6	0.4175	7	0.4589	6
Sport Climbing	0.5089	7	0.4294	5	0.3946	7
Breaking	0.46	8	0.4756	4	0.3052	10
Lacrosse	0.4517	9	0.424	6	0.4751	5
Coastal Rowing	0.3248	10	0.3744	8	0.3902	8
Flag Football	0.2528	11	0.1787	11	0.3278	9

6 Potential SDEs for Brisbane 2032 and Beyond

6.1 Potential SDEs for Brisbane 2032

We updated our SDE data for Brisbane 2032 and applied our validated H-MCDM model. Based on our model results shown in Table 7, the following three SDEs are recommended for inclusion in the Brisbane 2032 Olympics: **Skateboarding (4th in Brisbane)**, Sport Climbing (5th in Brisbane), and Breaking (6th in Brisbane) for their youth relevance.

Table 7: Scores and ranks of 11 selected SDEs evaluated for Brisbane 2032

SDE	Score	Rank
Athletics	0.8382	1
Gymnastics	0.6546	2
Swimming	0.6426	3
Skateboarding	0.6127	4
Breaking	0.4743	5
Sport Climbing	0.4457	6
Lacrosse	0.4378	7
Baseball/Softball	0.4141	8
Coastal Rowing	0.3828	9
Karate	0.359	10
Flag Football	0.1995	11

6.2 Olympic Inclusions for 2036 and beyond

Padel is a strong candidate for inclusion in the Olympics by 2036 or beyond as a result of its global growth, structured organization, and appeal to younger audiences. Originating in Mexico and highly popular across Europe and Latin America, padel has seen exponential growth. There are an estimated 25 million players worldwide as well as increasing professional leagues and tournaments, all governed by the International Padel Federation[50]. Its fast-paced, accessible gameplay combines elements of tennis and squash, creating an engaging spectator sport that aligns with the International Olympic Committee's (IOC) goal of attracting younger viewers. According to All for Padel[51], Padel has been given Association of IOC Recognized International Sport (ARISF) status and the only criteria it is lacking in is that it isn't played in enough countries yet. However, that should be easily reached by 2036 and beyond.
Pickleball is extremely likely to be included in the 2036 Olympics or beyond due to its global popularity, accessibility, and alignment with Olympic values of inclusivity and athletic skill. Originating in the U.S. but now rapidly spreading internationally, pickleball has experienced a 158.6% increase in participation from 2019 to 2023, making it the fastest-growing sport in the U.S. and one of the fastest-growing worldwide[52] . The sport's low cost, ease of

learning, and adaptability for various ages and skill levels make it highly inclusive, aligning well with the Olympic commitment to broadening accessibility and participation. In addition, the establishment of national and international governing bodies, such as the International Pickleball Federation (IPF), supports its potential Olympic pathway. However, the IPF is not on the list of IOC-recognized federations[53], possibly since it was created in 2023. As a result, all the sport needs is a little more time to keep developing.

VR-enhanced archery has potential for future Olympic inclusion, possibly as early as the late 21st century, as advancements in sports technology continue to blur the line between physical and digital athletics. VR-enhanced archery combines traditional archery skills with immersive technology. It demands precision, focus, and agility in a virtual setting—elements that align with Olympic standards for skill-based competition. This sport's appeal lies in its blend of physicality and virtual interactivity. This not only modernizes archery but also captures the attention of tech-savvy, younger audiences. According to channelnewsasia[54], the Olympics are starting to add "new and unconventional sports in the hope of appealing to younger people, who are thought to have markedly different tastes." As the IOC embraces innovation through events like the Olympic Esports Series, VR-enhanced sports like archery could become Olympic candidates, symbolizing the merger of tradition and technological evolution.

7 Sensitivity Analysis

Data in multi-criteria decision-making is often uncertain or subject to change. Therefore, sensitivity analysis is crucial for supporting accurate decisions and determining the reliability of results. We test the sensitivity of our model by adjusting the weights of the top-level criteria and analyzing their influence on the rankings of the alternatives. Specifically, we altered the weight of one criterion at a time up to 10% or 50%, while holding the others constant to identify the factors that most significantly influence the rankings. Due to the page limitation, we only presented our sensitivity analysis results using Tokyo as the host, but the results are consistent for other host cities.

7.1 Sensitivity Analysis: Impact of ±10% Change in AHP Weights

We adjust the weights of our top-level criteria by ±10% in 1% increments. Figure 5 shows how the scores of the sports change across this range. Popularity and Accessibility, along with Relevance and Innovation, have the greatest influence on rankings, aligning with the IOC's priorities. Criteria with lower weights, such as Gender Equity, Sustainability, and Safety and Fair Play, have minimal impact on the rankings. The model demonstrates stability, with Athletics, Gymnastics, and Swimming consistently dominating the top ranks. Youth-focused sports like Skateboarding, Sport Climbing, and Breaking show moderate changes in scores when criteria like Relevance and Innovation are adjusted, reflecting a clear connection to those factors rather than random shifts. Overall, the model is balanced and resilient to weight variations, showing it is robust.

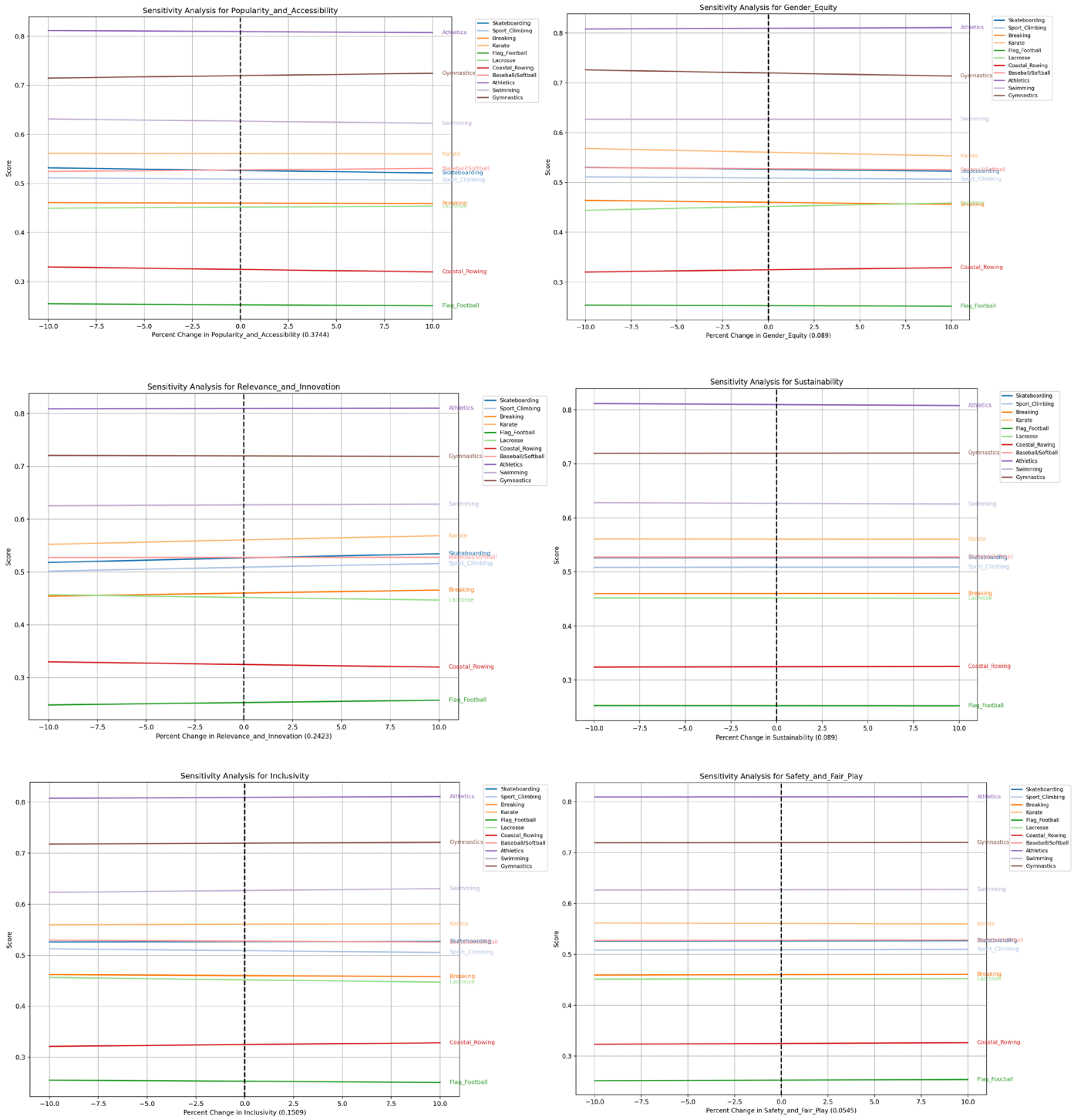


Figure 5: Impact of ±10% Weight Changes in the AHP weights on final score and ranking of 11 selected sports

7.2 Sensitivity Analysis: Impact of ±50% Change in AHP Weights

Next, we increase and decrease the weights of our top-level criteria by 50% in 1% increments in order to identify the criteria that changes our model the most. Our sensitivity analysis is displayed in the Figure 6 below:

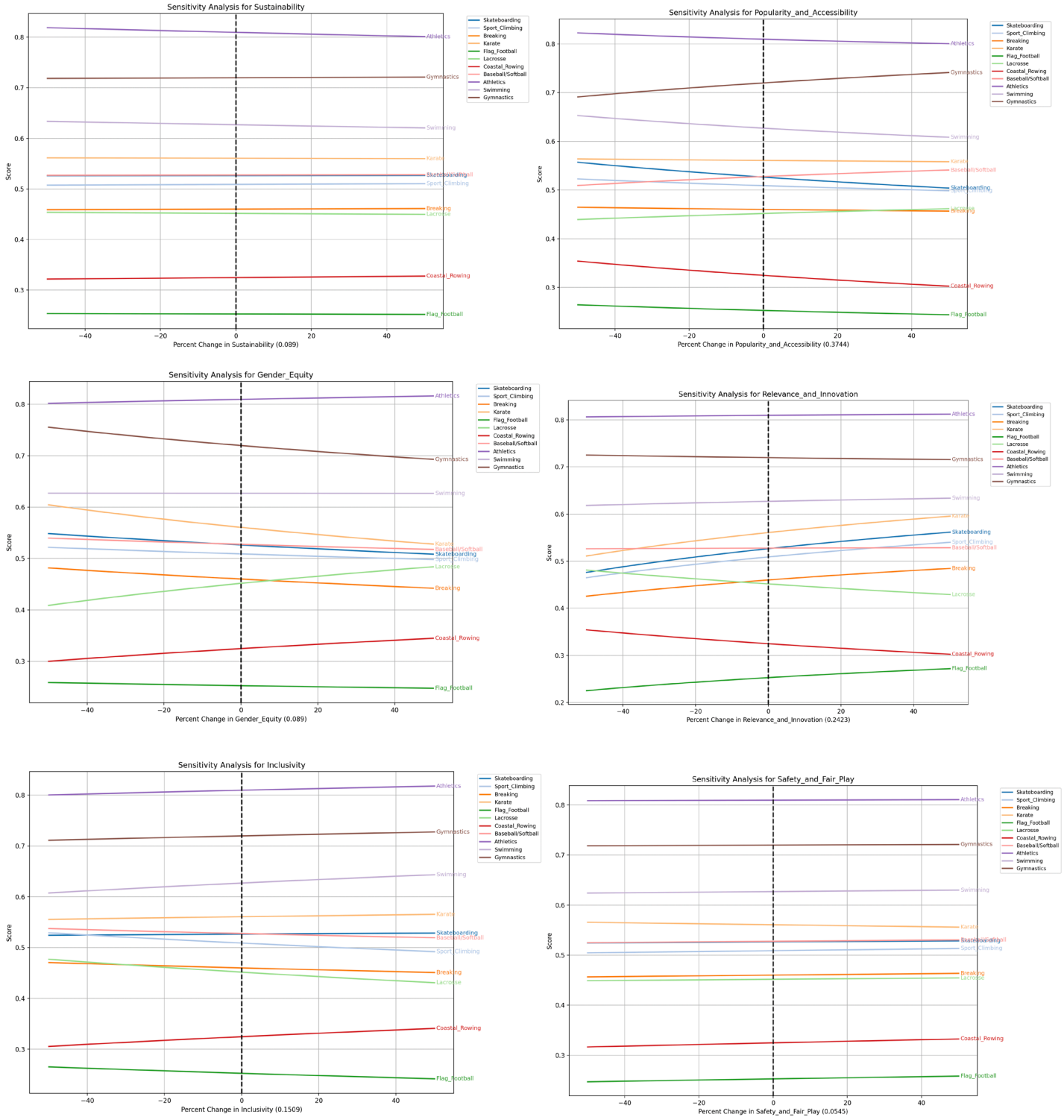


Figure 6: Impact of ±50% Weight Changes in the AHP weights on final score and ranking of 11 selected sports

The model remains stable, avoiding drastic or random ranking changes, even under ±50% weight adjustments. Athletics, Gymnastics, and Swimming consistently rank at the top, while Flag Football and Coastal Rowing stay at the bottom. Youth-focused sports like Skateboarding, Sport Climbing, and Breaking show meaningful responses to changes in criteria like Relevance and Innovation. **Popularity and Accessibility** and **Relevance and Innovation**

have the greatest impact on rankings, aligning with the IOC's focus on global appeal and youth engagement. **Gender Equity** and **Inclusivity** show a moderate influence, while **Sustainability** and **Safety and Fair Play** have minimal effect. Overall, the model effectively captures key trends without causing drastic or random ranking changes.

7.3 Model Strengths and Weaknesses

The Hierarchical Multi-Criteria Decision Model effectively integrates AHP and TOPSIS within a hierarchical framework to evaluate and rank SDEs for Olympic inclusion. By systematically considering both the relative importance of criteria (through AHP) and the performance of each SDE against these criteria (through TOPSIS), the model provides a robust and transparent tool for decision-makers. This approach ensures that the selected SDEs align closely with the IOC's evolving vision and strategic objectives. More specifically, we listed our model's strengths and weaknesses below.

7.3.1 Strengths

- **Structured Evaluation:** Reflects the multi-level nature of the IOC's criteria, ensuring that both high-level and detailed factors are considered.
- **Comprehensive Evaluation:** By combining multiple criteria, the model ensures a balanced and holistic assessment of sports.
- **Flexibility:** Can accommodate changes in criteria or the addition of new sub-criteria without significantly impacting the entire model.
- **Transparency:** Each step of the evaluation is clear and can be examined individually, aiding in understanding and acceptance of the results.
- **Stability:** Sensitivity analysis shows that the model remains stable with changes in criteria weights, ensuring reliability in rankings.

7.3.2 Weaknesses

- **Dependence on Data Quality:** The accuracy of rankings heavily depends on the availability and reliability of input data.
- **Underrepresentation of Low-Weighted Criteria**: Lower-weighted criteria like Sustainability and Safety and Fair Play may have limited influence on rankings as shown in our sensitivity analysis.
- **Subjectivity in Weight Assignment:** While AHP helps determine weights, the initial pairwise comparisons rely on subjective judgments, which may introduce bias. Also, pre-defined weights for sub-criteria introduce subjective biases that may affect the outcomes.

8 Conclusion

In order to help the IOC quantitatively evaluate which SDEs align best with Olympic criteria for optimal selection, we proposed a model, the Hierarchical Multi-Criteria Decision Model, to generate scores for any SDE, justifying their removal or inclusion by evaluating each SDE against the IOC Criteria for Sports Inclusion. We used 2-6 quantitative and/or qualitative factors for each of the 6 IOC criteria, adding up to 27 total factors. Some of these factors depend on the host city, reflecting the Olympics' evolving vision.

Using historical data about these 27 factors, our proposed model integrates both AHP and TOPSIS to determine the final host-city specific score of each SDE. Our final scores range from 0-1, where 1 represents the highest

alignment with the IOC criteria among the evaluated SDEs, and 0 represents the lowest alignment. SDEs will be ranked from highest to lowest based on these scores. We chose 11 SDEs, eight of which were recently added or removed, and three core Olympic sports. We used our model to rank each one with respect to different host cities. We first applied our model to Olympics that have already passed or has determined SDEs, specifically Tokyo 2020, Paris 2024 and LA 2028, in order to justify each SDE's removal or inclusion in those Olympics. The validity of our model is confirmed through the rankings of sports that are not stable and instead host-specific. For example, karate was included in the Tokyo 2020, but was taken out for Paris 2024 and LA 2028. Our model assigned karate a relatively high score of 0.5607 and a high rank of number four for Tokyo 2020. However, for Paris 2024 and LA 2028, the model outputted scores of 0.3369 and 0.2621 with ranks of #10 and #11, respectively. These results were confirmed by the status of karate in these three Olympics. In addition, breaking was included in Paris 2024 with a score of 0.4756 and a rank of #4, however for LA 2028, the model gave a score of 0.3052 and a rank of #10, consistent with the fact that Breaking is removed for LA 2028. For baseball/softball, our model ranked it high (rank of #4) for LA 2028, while giving a much lower rank for Paris 2024 (rank of #9). Ultimately, the recently added sports have a higher rank when compared to those that are recently removed, suggesting the validity of our model.

We then applied our validated model to evaluate the eight recently added or removed SDEs for Brisbane 2032, changing the data with respect to the host city. Our model predicts that the first SDE to be considered for inclusion is Skateboarding with a score of 0.6127, the second is Breaking with a score of 0.4743, and the third is Sport Climbing with a score of 0.4457. For the Olympics 2036 or beyond, we believe that pickleball, padel and VR-archery are possible inclusions. The key reason is that these three sports fit the trends that the younger generation supports, helping the IOC fulfill the criteria of relevance and innovation.

A sensitivity analysis was performed to test the robustness of our weights, as we changed the weight number slightly, analyzing the difference it made to final ranking of 11 chosen SDEs. We found that our model produces robust rankings for the 11 SDEs when the weight is changed slightly. In addition, Popularity and Accessibility, along with Relevance and Innovation, have the greatest influence on rankings, aligning with the IOC's priorities.

In summary, our proposed model effectively adapts to varying cultural, regional, and strategic priorities of different hosts. Our model's criteria and weighting approach is validated through the alignment of high-ranked sports with actual Olympic decisions (Tokyo 2020, Paris 2024, LA 2028) and hence our model provided solid quantitative reasoning for those three sports recommended to be considered for the 2032 Olympics.

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Report on Use of AI

We used ChatGPT for multiple tasks during the competition:

1. **Code development of our Hierarchical Multi-Criteria Decision Model (H-MCDM)**

   We used ChatGPT (version o1-preview or 4.0 with Canvas) to assist in developing a Python code for AHP, TOPSIS, and H-MCDM. Additionally, ChatGPT helped us create a code for sensitivity analysis and generate relevant plots. We thoroughly understood, verified, and tested all generated code to ensure accuracy. Developing our solution involved a lot of interactions with GPT, which made the process highly efficient and resulted in excellent code accuracy. A sample query is provided below:

   Sample Query:
   "For all fields, the following columns: Infrastructure_Cost, Athlete_Quota, infrastructure_cost_Availability_of_Venues, Injury_Rate_per1000, are 'min-best'. Could you update the code to check whether a column is one of these 'min-best', and if so, apply 'min-best' positivization?"

   Sample Output:
   "I've updated the Positivization function to automatically apply 'min-best' positivization for specific columns (Infrastructure_Cost, Athlete_Quota, infrastructure_cost_Availability_of_Venues, Injury_Rate_per1000). Let me know if you need further adjustments or additional modifications."

2. **Looking for data for our factors**

   Collecting data was very challenging during our math modeling, which involved 11 SDEs and 27 factors. The current version of GPT (GPT-4.0) has proven quite effective in providing information and suggesting data sources. While we understand that GPT can sometimes provide incorrect or non-existent information, especially in earlier versions, our experience with GPT-4.0 has been very positive. For every suggestion it provided, we verified the data by consulting multiple sources. In our experience, if the prompt is highly specific and asks detailed questions, GPT delivers fairly accurate and reliable information. Although it occasionally makes mistakes, the majority of the data and sources it provided were accurate.

   Sample Query:
   "How is the Leadership Gender Balance of the World Karate Federation (WKF)?"

   Sample Output:
   "As of November 12, 2024, the World Karate Federation (WKF) Executive Committee comprises 23 members, including the President, Vice Presidents, and other key officials. Among these, 3 are women, resulting in a female representation of approximately 13%. This indicates a gender imbalance within the leadership structure, with male members constituting the majority."

The response also provided 13 citation links, as shown in Figure 7, which we reviewed one by one and often explored additional links during our reading. After this verification process, we were confident that GPT's answer was accurate.

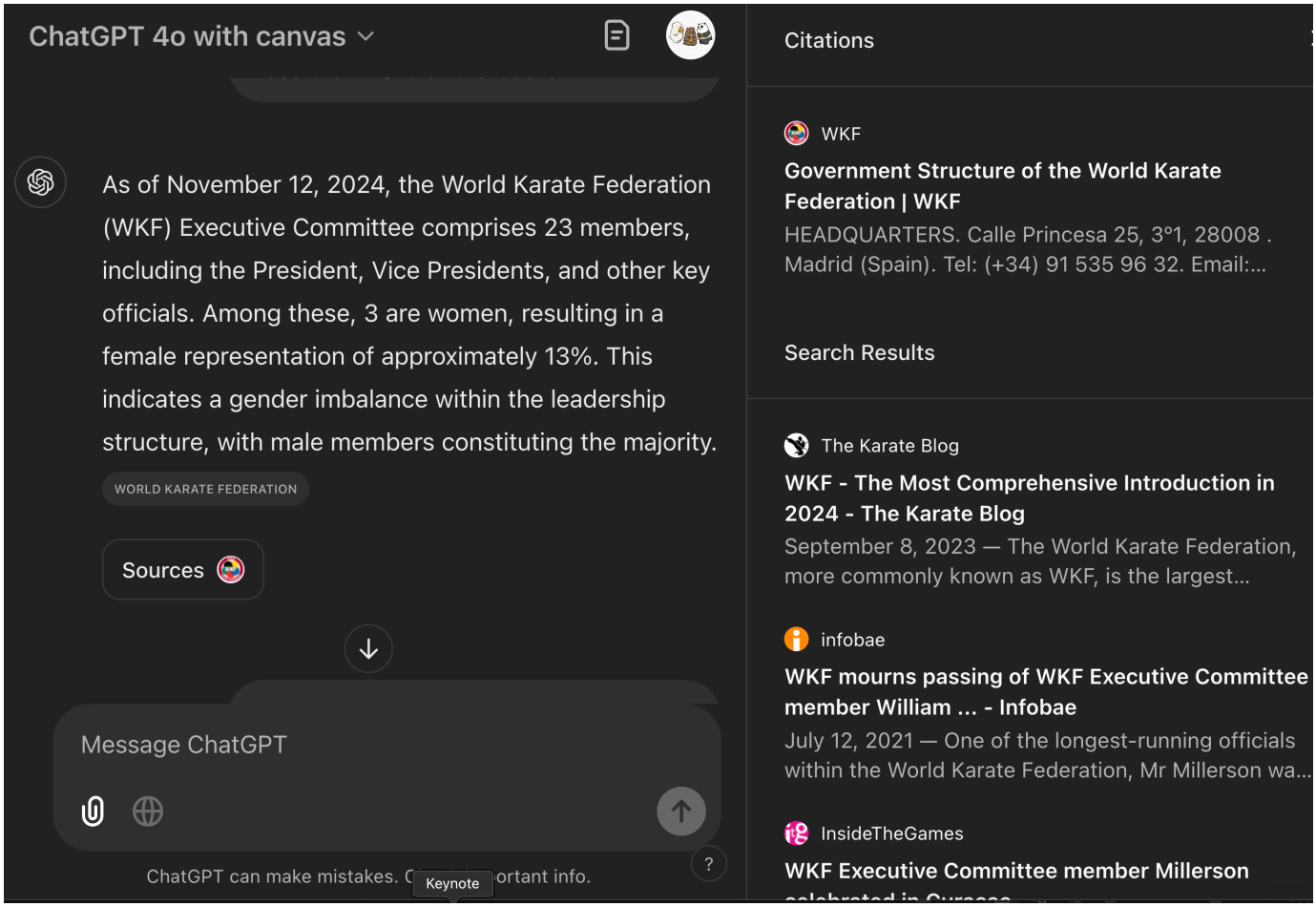


Figure 7: Sample ChatGPT snapshot showing the response to the query, 'How is the Leadership Gender Balance of the World Karate Federation (WKF)?'

3. **Editing Assistance and Grammar Correction**

GPT-4.0 with Canvas proved to be an excellent tool for improving our writing and checking for grammatical errors. After drafting a paragraph, we used GPT-4.0 with Canvas to evaluate our writing and identify any grammar issues. It was particularly effective at detecting flaws in text flow and pointing out grammatical mistakes. After completing an entire section, we also asked for its suggestions for improvement. GPT-4.0 with Canvas provided insightful feedback on clarity, structure, and coherence, helping us enhance the overall quality of our writing. It offered detailed recommendations on reorganizing content, refining arguments, and improving transitions between ideas, resulting in a polished version.

Sample Query:

Please note that several factors are evaluated using a 'min-better' approach, meaning that lower values are preferred. Specifically, Infrastructure Cost, Athlete Quota, and Athlete Health and Safety - Injury Rate are considered 'min-better' as lower values indicate enhanced feasibility or safety. Additionally, for Equal Events and Participation, a ratio of 1:1 is considered optimal, representing a balanced middle point. All other factors (the majority of factors) follow a 'max-better' approach, where higher values are preferred.

Sample Output:

Here's a slightly revised version with a simplified tone:

Please note that some factors are evaluated using a 'min-better' approach, where lower values are preferred. Specifically, **Infrastructure Cost**, **Athlete Quota**, and **Athlete Health and Safety - Injury Rate** fall into this category, as lower values indicate greater feasibility or safety. For **Equal Events and Participation**, an optimal ratio is 1:1, representing a balanced midpoint. Most other factors use a 'max-better' approach, where higher values are desirable.

4. Math Modeling Brainstorm and Ideas

We initially consulted GPT-4 for modeling ideas and suggestions, and it provided some useful information. However, after conducting extensive research through Google, we felt more confident relying on our own findings rather than depending solely on GPT. While GPT offered a helpful starting point, deeper exploration on the internet led to more creative ideas and more specific, valuable information.